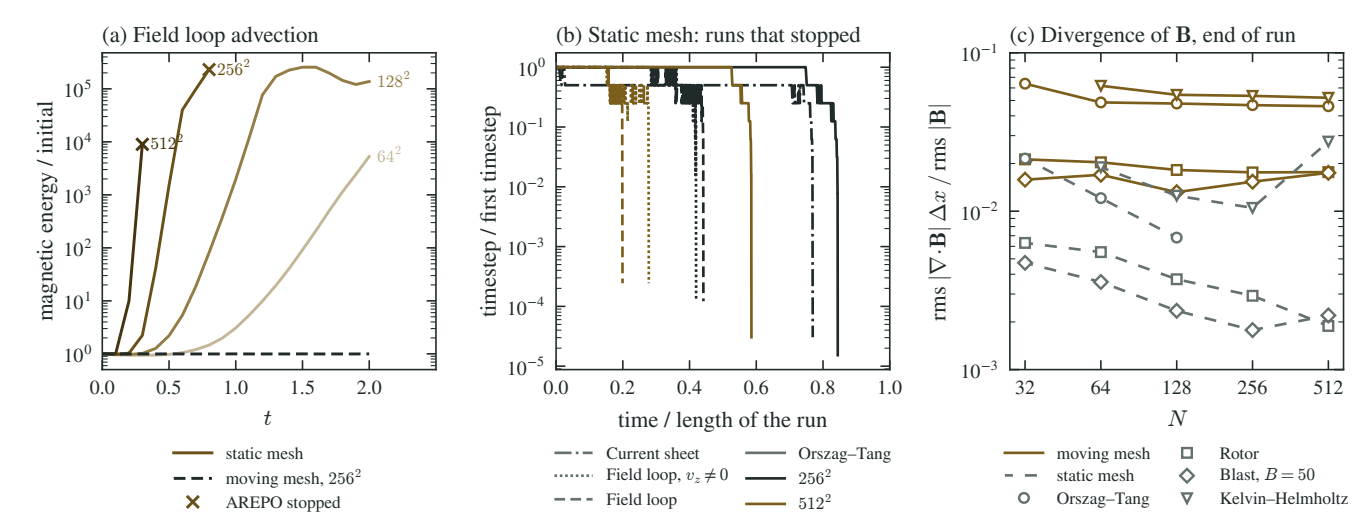
(a) Field loop advection
magnetic energy / initial
10^0
10^1
10^2
10^3
10^4
10^5
0.0
0.5
1.0
1.5
2.0
t
256^2
512^2
128^2
64^2
static mesh
moving mesh, 256^2
AREPO stopped
(b) Static mesh: runs that stopped
timestep / first timestep
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0.0
0.2
0.4
0.6
0.8
1.0
time / length of the run
Current sheet
Field loop, $v_z \neq 0$
Field loop
Orszag–Tang
256^2
512^2
(c) Divergence of B, end of run
rms $|\nabla \cdot \mathbf{B}|\,\Delta x$ / rms $|\mathbf{B}|$
10^{-1}
10^{-2}
10^{-3}
32
64
128
256
512
N
moving mesh
static mesh
Orszag–Tang
Rotor
Blast, $B = 50$
Kelvin–Helmholtz